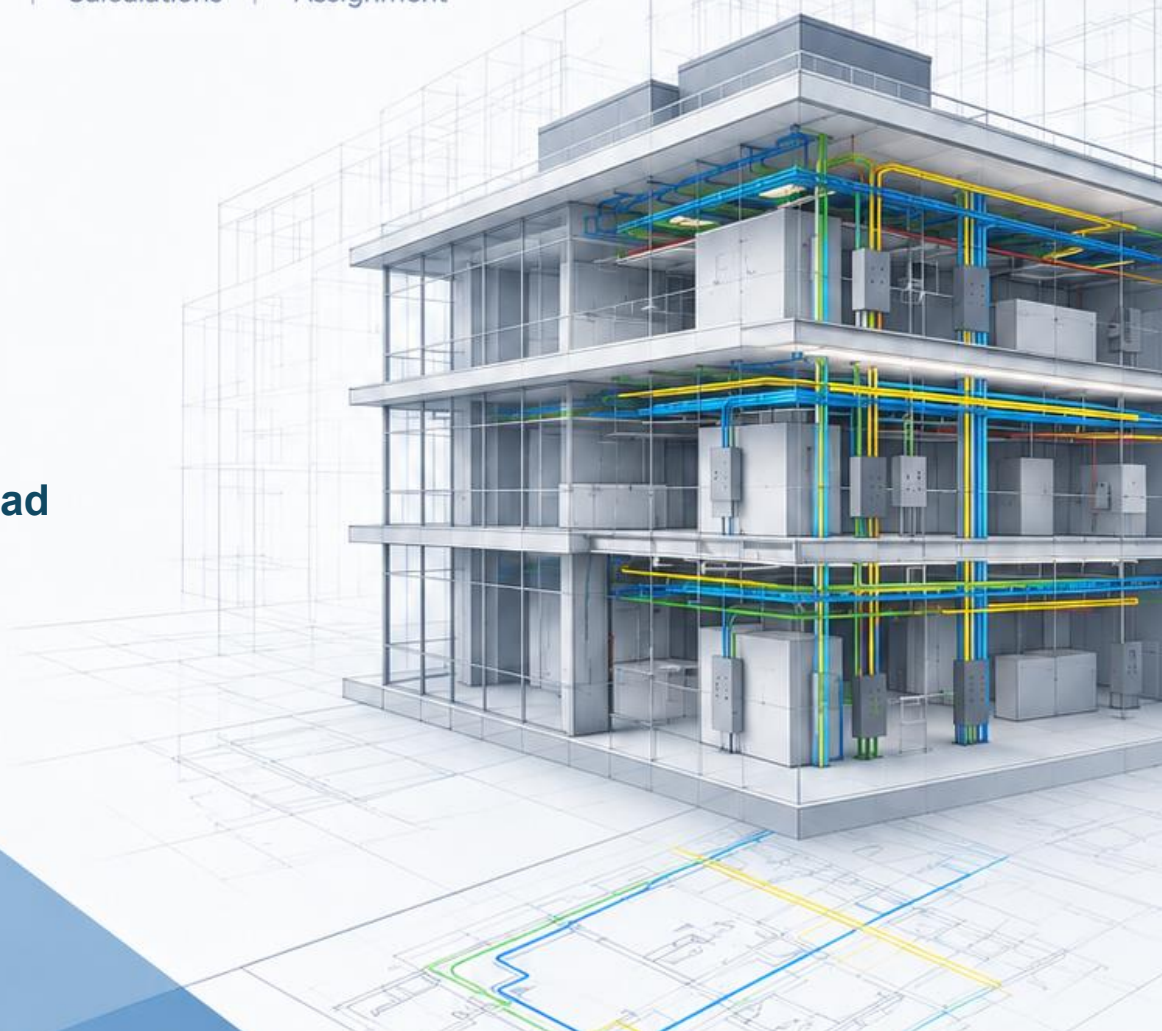
Cable Engineering Tool

Revit Add-in

Software Architecture Report

System Architecture & Design Overview

Revit Integration | Cable Design | Electrical Calculations | Reporting & Assignment



Version 1.0

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1. Introduction

The Cable Engineering Tool is a Revit add-in developed to support electrical cable engineering workflows according to IEC standard by integrating electrical design calculations, cable routing, project validation, and data management within the Autodesk Revit environment.

The tool combines Revit model data with dedicated calculation and routing engines to support cable sizing, voltage drop calculations, short-circuit verification, circuit breaker selection, neutral conductor and harmonic assessment, and earthing conductor sizing for TN-S and TT systems.

A graph-based routing approach using Dijkstra's algorithm is used to determine cable paths through available electrical containment elements. The resulting route information supports engineering calculations that depend on the actual cable path length.

The application also provides project validation capabilities to identify relevant model data issues before dependent operations are performed, presents calculation results through a Windows Presentation Foundation (WPF) interface, and supports exporting engineering data and results to Excel.

This document describes the software architecture, principal components, data flow, Revit integration, and key design decisions underlying the tool.

1.1 Purpose and Scope

The purpose of this document is to describe the software architecture and design of the Cable Design Engineering Tool, providing a structured technical reference for understanding how its components interact and how engineering workflows are implemented within the Revit environment.

The document aims to:

- Describe the system architecture, major software components, and their responsibilities.
- Explain the flow of data from Revit model extraction and project validation through routing, electrical calculations, and result presentation.
- Document the integration between the Revit API, the WPF user interface, calculation engines, and reporting functionality.
- Explain key architectural decisions and the rationale behind the separation of application responsibilities.
- Establish a technical reference that supports future maintenance, testing, and controlled development.

This document covers the architecture and implementation of the current version of the Cable Engineering Tool, including:

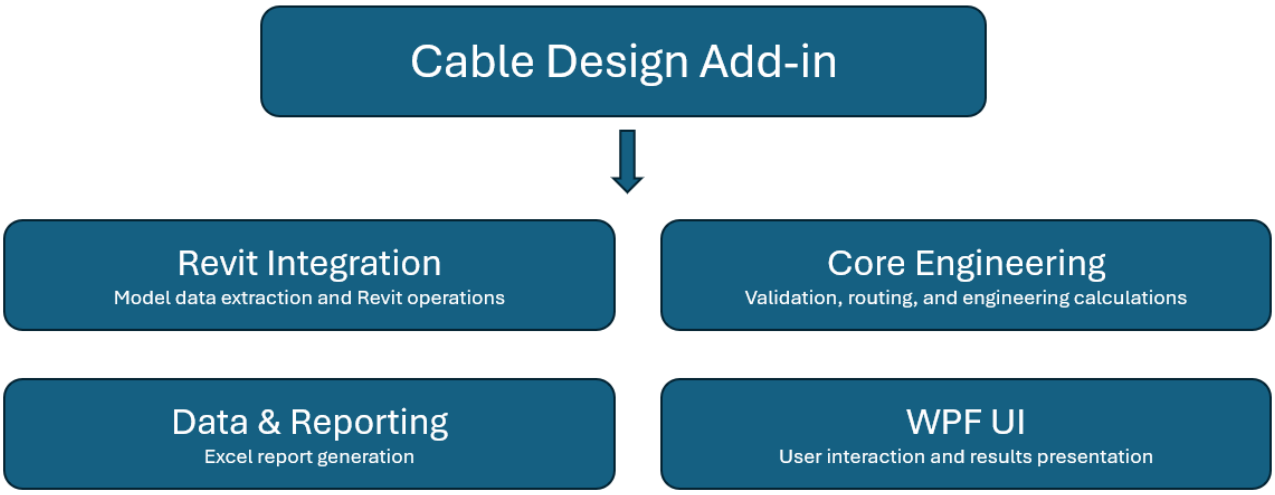
- Revit model data extraction and project validation.
- Cable routing and route-length determination.
- Electrical calculation engines and their interaction with project data.
- WPF-based user interface and application workflow.
- Cable assignment and Revit model updates.
- Excel reporting and data export.
- Performance considerations, error handling, testing, and current system limitations.

The document focuses on software structure and implementation. Detailed electrical equations, applicable standards and numerical verification results are addressed separately in the Cable Engineering Calculation Methodology and Verification Report.

The scope of supported workflows and system configurations is limited to the functionality implemented and verified in the current release. Planned enhancements are identified separately as future development items.

2.System Architecture Overview

2.1 High-Level Architecture Diagram



2.2 Component Responsibilities and Interactions

COMPONENT	RESPONSIBILITIES	INTERACTIONS
REVIT INTEGRATION	Extracts required data from the Revit model and executes Revit-specific operations.	Provides model data to the Core Engineering component and executes Revit operations requested by the application and assign cables to Revit.
CORE ENGINEERING	Performs project validation, circuit routing, and electrical engineering calculations.	Processes project data received from Revit Integration and provides validation and calculation results to the application and reporting components.
REPORTING	Manages the preparation and export of project validation and engineering results to Excel reports.	Retrieves the required project data and results from the calculations to generate reports.
WPF UI	Presents application pages, collects user inputs, and displays validation status, routing information, and calculation results.	Communicates with the Core Engineering and Revit Integration components to request processing, retrieve results, and initiate Revit operations.

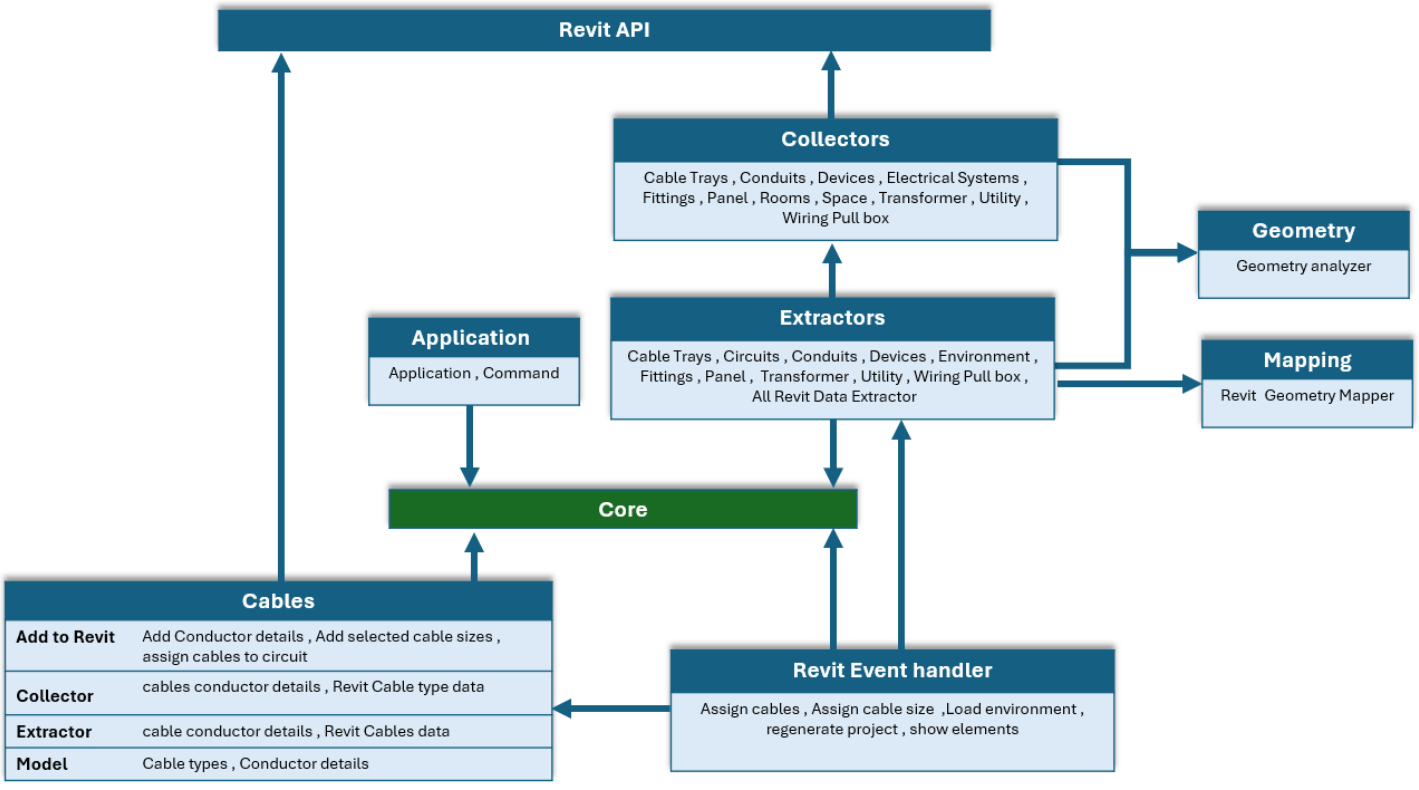
3. Application Structure and Components

3.1 Revit Integration

The Revit Integration layer connects the Cable Engineering Tool with the Revit environment and provides access to the Revit API. It is responsible for collecting the required project data, extracting and translating it into the application's core data models, handling Revit-specific operations, and supporting cable assignment to Revit circuits.

This layer is organized into several components, including Collectors, Extractors, Geometry, Mapping, the Revit Event Handler, and Cables. Each component has a specific responsibility and interacts with other components according to the application's implementation.

The following diagram illustrates the main components of the Revit Integration layer and their dependencies, providing an overview of how they are organized and how they interact within the application architecture.

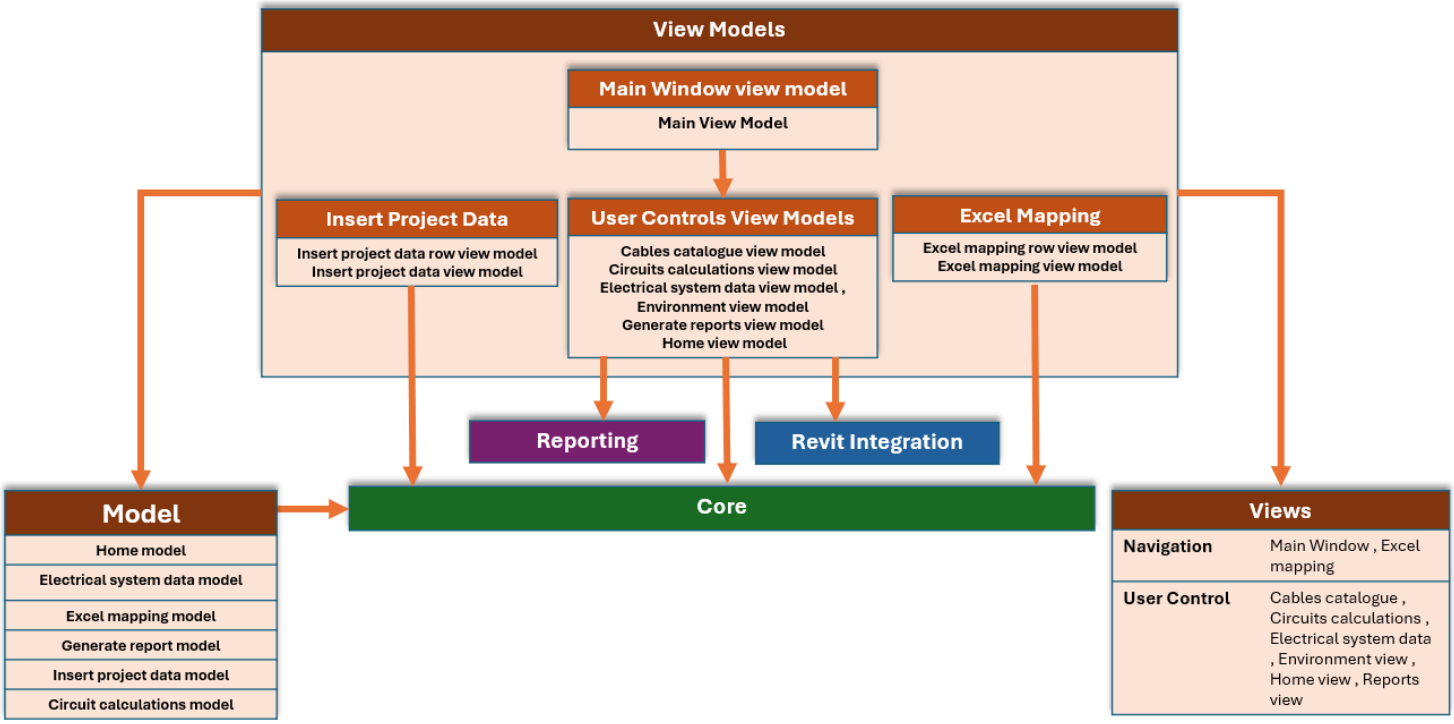


COMPONENT	RESPONSIBILITIES	INTERACTIONS
COLLECTORS	Collect the required Revit element data from the active Revit project using the Revit API.	**Revit API**: Retrieves elements and their properties. **Extractors**: Provides the collected data to the extractors that request it.
EXTRACTORS	Extract and transform collected Revit data into the corresponding Core models required by the application. Coordinate the individual extraction processes through All Revit Data Extractor.	**Collectors**: Requests the required Revit data. **Core**: Uses Core models to represent the extracted data. **Geometry** / **Mapping**: Uses their helper functions when required by the extraction process.
APPLICATION	Provides the add-in entry point within Revit, exposes the add-in through the Revit UI, and launches the project command to initiate the workflow.	**Revit UI**: Integrates the add-in into Revit. Project **Command**: Initiates the project workflow. **WPF UI**: Opens the tool interface, if launched by the command.
GEOMETRY	Converts Revit API XYZ locations and coordinates into the corresponding Core geometry representation.	Revit API: Reads Revit geometry and coordinates. Core: Uses the corresponding Core coordinate type. Extractors / Mapping: Provides geometry conversion when called by these components.
MAPPING	Maps Revit environment location information to the corresponding environment data used by the application.	**Revit environment data**: Uses the relevant location information. **Core:** Maps the information to the appropriate Core environment representation. **Geometry:** Uses coordinate conversion where required.
REVIT EVENT HANDLER	Executes Revit-dependent operations requested by the UI through Revit's event mechanism and coordinates the required extraction or cable-related operations. Makes operation results available to the UI through the application's existing result or notification mechanism.	**WPF UI**: Receives requests raised through the external-event mechanism. **Revit API:** Executes operations in a valid Revit API context. **Extractors / Core / Cables**: Coordinates the relevant operations, where directly used by the handler. **UI notification mechanism**: Makes results available to the interface.
CABLES	Handles cable-related operations in Revit, including applying imported cable information and assigning selected cable data to the appropriate Revit circuits.	**Revit API:** Accesses and updates the relevant Revit cable and circuit information**.** **Imported cable data**: Uses the selected cable information. **Revit Event Handler:** Executes Revit operations through the event mechanism, if invoked by the handler.

3.2 WPF UI Architecture

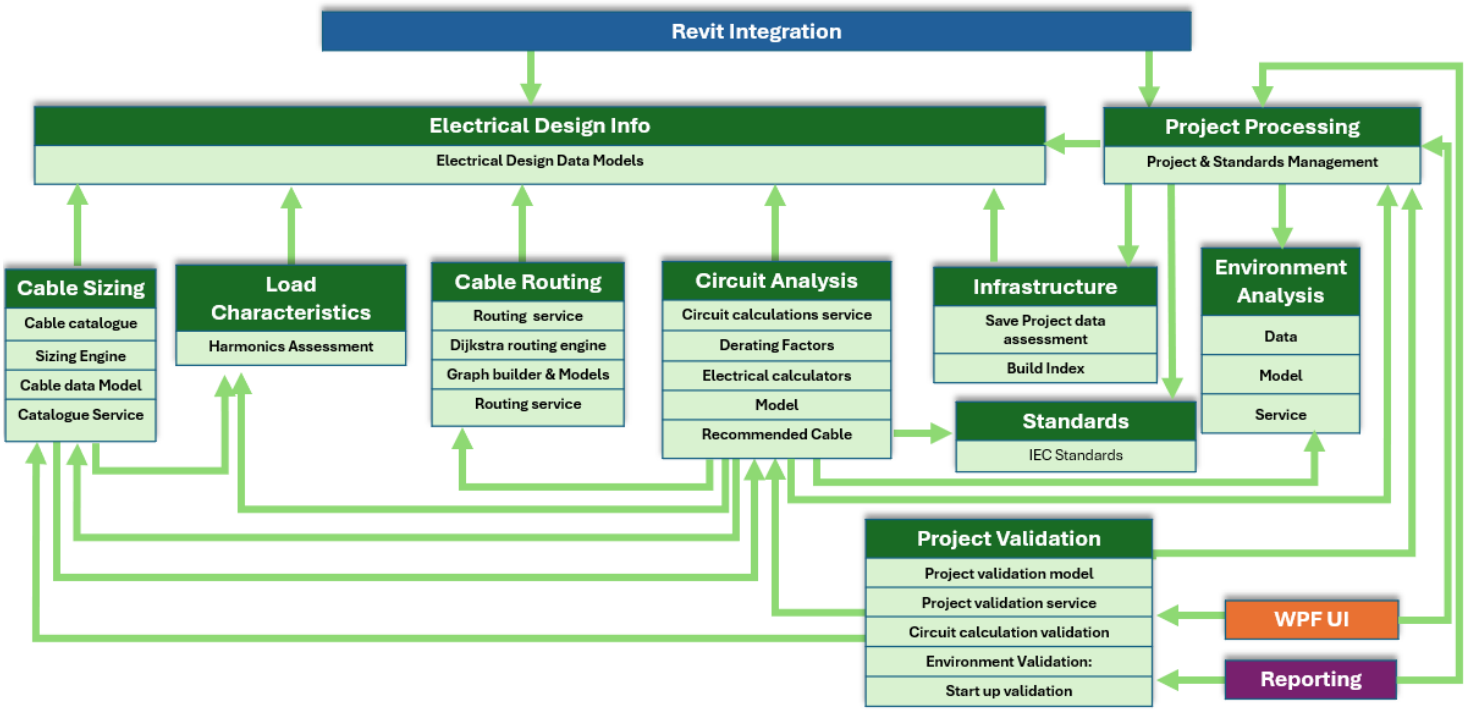
The WPF UI layer provides the user interface for interacting with the Cable Engineering Tool. It follows the MVVM pattern, separating data presentation, UI state management, and user interactions. The layer consists of Models, Views, and View Models, which work together to display Revit data, present calculation results, and coordinate user-initiated operations with the Core and Revit Integration components.

The following diagram illustrates the main components of the WPF UI layer and their dependencies.



COMPONENT	RESPONSIBILITIES	INTERACTIONS
MODELS	Represent and hold the data objects used to display and manage information in the UI.	Used by View Models to access, update, and present data through data binding.
VIEWMODELS	Manage UI state and commands, coordinate workflows, and prepare data for presentation. Handle user requests and coordinate operations involving Revit data and Core services.	Interact with Models, Core services, Project Context, and Revit handlers. Provide data and commands to Views through data binding and command binding.
VIEWS	Display Revit data, present calculation results, receive user input, and provide controls for initiating operations such as assigning cables to circuits and exporting reports.	Bind to View Models to display data and execute commands. May use view-specific methods for UI presentation tasks.

3.3 Core Engineering

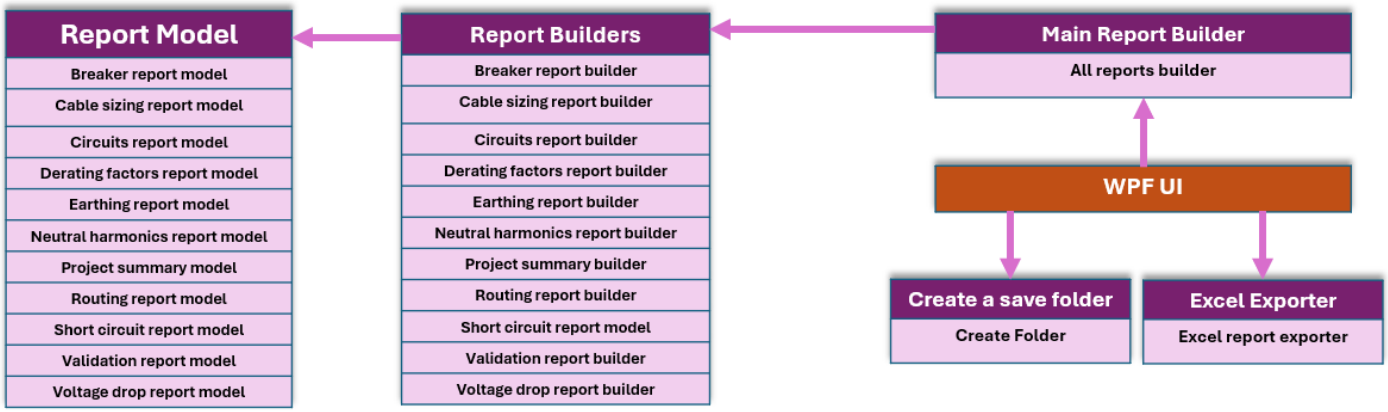


COMPONENT	RESPONSIBILITIES	INTERACTIONS
ELECTRICAL DESIGN DATA MODELS	Represent and hold the data objects used to manage electrical design information within the Core.	Provide the data structures consumed and updated by the Core services and calculation engines.
CABLE SIZING	Provide cable catalogue management, cable sizing engines, and related sizing models.	Uses cable data and IEC standards to perform phase, neutral, and earthing cable sizing.
LOAD CHARACTERISTICS	Manage harmonic data and provide harmonic assessment services for electrical calculations.	Supplies harmonic assessment results to the relevant calculation engines, particularly neutral sizing.
CIRCUIT ANALYSIS	Perform electrical analysis of circuits, including voltage drop, short-circuit protection, neutral sizing, and earthing calculations.	Uses circuit data, cable sizing results, routing information, load characteristics, and IEC standards to evaluate circuit performance and determine suitable cable configurations.
CABLE ROUTING	Determine the cable route for each circuit using graph-based routing and Dijkstra's algorithm.	Uses the project infrastructure and routing graph to calculate routes, applying routing-

		type filters and providing route results to the subsequent cable calculation workflows.
INFRASTRUCTURE	Manage project data persistence in JSON files and maintain project indexes for efficient element lookup.	Provides project data and indexed element access to routing, validation, and calculation components.
ENVIRONMENT ANALYSIS	Determine cable installation environmental conditions and associated temperature correction factors.	Uses project environment data and installation information to provide temperature-related inputs to cable sizing calculations.
STANDARDS	Store and provide IEC standard data, including tables and parameters required for electrical calculations.	Supplies standard values and calculation parameters to cable sizing, circuit analysis, and validation components as required.
PROJECT VALIDATION	Validate project data, installation environment, and calculation inputs before they are used by the UI and calculation workflows.	Uses project data, environment information, and calculation requirements to produce validation results for the UI and other dependent components.
PROJECT PROCESSING	Manage project and standards contexts and load the required project data, environment settings, and standards configuration.	Provides the loaded project context and standards context to the relevant Core components for validation, routing, and electrical calculations.

3.4 Reporting

The reporting architecture is designed to generate and export the tool's reports through a structured and coordinated process. It consists of report models, individual report builders, a main report builder, and an Excel exporter, with the WPF View Model initiating the process and handling the user interaction.



COMPONENT	RESPONSIBILITIES	INTERACTIONS
REPORT MODELS	Represent and hold the data objects required to generate the reports.	Used by the Report Builders to structure report data.
REPORT BUILDERS	Generate report content for the different sections of the tool.	Use Report Models and receive the required data from the report-generation View Model.
MAIN REPORT BUILDER	Coordinate all Report Builders and assemble the complete report content through a single method.	Uses the individual Report Builders and their outputs to produce the complete report for the WPF View Model.
CREATE SAVE FOLDER	Allow the user to select the destination folder for the Excel report.	Provides the selected folder path to the Excel Exporter.
EXCEL EXPORTER	Export the complete report into an Excel file at the specified path.	Receives the assembled report content and the destination file path.

4. Engineering Workflows

4.1 Routing workflow

- Build the routing graph from the project infrastructure.
- Filter routing elements according to the routing type.
- Calculate the shortest route for each eligible circuit using Dijkstra's algorithm.
- Store the calculated route and its actual length for subsequent calculations

4.2 Calculation Workflow

- Determine the required cable capacity based on the circuit load and applicable correction factors.
- Select suitable cables from the imported cable catalogue.
- Calculate voltage drop and short-circuit withstand capacity.
- Calculate all Derating factors.
- Perform neutral conductor, Harmonics and earthing calculations.
- Evaluate the selected cable against the applicable design requirements and standards.

4.3 Environment workflow

- Determine the installation environment for each cable route in the project.
- Retrieve the ambient temperature and reference temperature associated with each route.
- Allow the user to review and modify the ambient temperature assigned to each route.

4.4 Validation Workflow

- Validate project readiness and required Revit data.
- Check the environment configuration and routing infrastructure.
- Validate the inputs required for electrical calculations.
- Identify invalid or incomplete circuits and report validation issues.
- Determine which circuits can proceed through the engineering workflow.

4.5 Revit Assignment Workflow

- Identify the circuits eligible for cable assignment.
- Associate the selected cable with the corresponding circuit.
- Assign the cable information to the relevant Revit electrical elements and parameters.
- Report the assignment results and any circuits that could not be processed.

5 .Data Management and Communication

5.1 Data Sources

- **Revit Model:** Electrical circuits, panels, cable trays, conduits, fittings, devices, utilities, and transformers.
- **Excel Cable Catalogue:** Cable specifications imported by the user.
- **JSON Files**: IEC standards data, harmonics library, environment settings, and saved project information.

5.2 Flow between Components

- Revit data is extracted and mapped into Core data models.
- Project Processing loads the project context and standards context.
- Project Validation checks the required project data and identifies issues.
- Cable Routing generates circuit routes, which are used by the calculation components.
- Circuit Analysis uses circuit data, cable catalogue data, environmental conditions, and applicable standards to calculate the results.
- The reporting components collect the results and export them to Excel.
- The Revit Assignment workflow writes the selected cable information back to the Revit model.

6. Design Decisions and Constraints

6.1 Separation of Responsibilities

The tool separates the WPF user interface, Revit integration, and Core engineering logic. This separation keeps Revit-specific operations independent from the electrical calculation engines and allows each component to be maintained and developed separately.

6.2 Version 1 Limitations

Version 1 focuses on the core cable design workflow, including project validation, cable routing, electrical calculations, reporting, and cable assignment to Revit circuits. Some capabilities are intentionally limited to the initial release scope, such as supporting one earthing system configuration (TT or TN-S) per project.

6.3 Architecture Limitations

The current architecture is designed around Revit as the source of project geometry and electrical system data. The tool operates as a Revit add-in and depends on the availability and validity of the required model data. Its current integration and processing approach may also affect performance when handling large projects.